

Alexa Petrovskaya

Software Engineering Intern

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EXPERIENCE

Stripe | Summer 2023 (3 months)

Software Engineering Intern

- Optimized payment processing pipeline by refactoring legacy fraud detection service; reduced average latency from 287ms to 156ms (46% improvement) affecting 850K+ daily transactions
- Implemented distributed caching layer for card validation using Redis; improved throughput from 4.2K to 18.5K requests/sec (340% gain) and reduced database load by 67%
- Shipped 8 pull requests merged by senior engineers; wrote comprehensive unit tests achieving 94% code coverage and prevented 3 critical race conditions before production deployment

Figma | Summer 2024 (3.5 months)

Frontend Engineering Intern

- Built real-time collaboration feature for shape drawing; optimized WebSocket message batching reduced network payload by 73% and improved sync latency from 420ms to 89ms for 12K+ concurrent users
- Refactored canvas rendering pipeline to use WebGL instead of Canvas2D; achieved 3.8x faster frame rendering (from 24fps to 91fps) on 4K viewport with 50K+ elements
- Conducted accessibility audit fixing 27 WCAG 2.1 AA violations; implemented keyboard navigation improving keyboard-only user coverage from 41% to 94% across design editor

Wise (formerly TransferWise) | Winter 2024 (2 months)

Full-Stack Engineering Intern

- Developed internal analytics dashboard using React + TypeScript + GraphQL; reduced report generation time from 6.2 seconds to 0.8 seconds (7.75x speedup) for 500K+ transaction queries
- Migrated legacy payment gateway integration from REST to gRPC protocol; achieved 42% reduction in latency (340ms → 197ms) and 58% lower bandwidth consumption across 200+ daily payment routes
- Fixed 12 production bugs identified through monitoring alerts; reduced API error rate for money transfer endpoints from 0.34% to 0.07% and saved \$47K in failed transaction refunds

PROJECTS

QueryVault | 2023 - Present

Rust, PostgreSQL, Tokio, Protocol Buffers

- Built high-performance distributed query cache middleware; 2,400+ GitHub stars, benchmarks show 8.9x latency reduction for repeated queries (from 340ms to 38ms) and handles 45K QPS at p99<50ms
- Implemented consensus protocol for cache invalidation across 7-node cluster achieving <200ms consistency window; 2.3K+ releases on crates.io with 89K+ monthly downloads

PixelForge | 2023 - 2024

TypeScript, Canvas API, Vite, Jest

- Open-source pixel art editor with collaborative features; reached 1,680+ GitHub stars in 8 months, 890+ active monthly users, and featured in ProductHunt #3 trending
- Optimized rendering performance from 15fps to 60fps+ on large canvases (8K×8K pixels) through WebWorker architecture; reduced initial bundle size from 2.4MB to 380KB via tree-shaking

SyntaxShield | 2024

Python, LLVM, AST parsing, pytest

- Static analysis linter for Python detecting 34 categories of code smells and security issues; 476 GitHub stars with 12K+ PyPI downloads since launch 2 months ago
- Achieves sub-second analysis on projects up to 100K lines of code; correctly identifies 97.2% of intentional vulnerabilities in test suite and integrates with 8+ popular IDEs

EDUCATION

Imperial College London | 2022 - Present

BSc Computing (2nd Year)

- Achieved 78% in Year 1 (equivalent to 1st class honors track); ranked top 12% in Algorithms & Data Structures module with 89%
- Led peer study group of 18 students in discrete mathematics, improving cohort average from 71% to 76% over 1 semester

University of Amsterdam | 2023 - 2024

Exchange Year - Computer Science

- Completed advanced systems programming course with distinction (9.2/10); built a kernel module achieving 12% faster I/O throughput
- Participated in ACM ICPC qualifying round, solving 4/10 problems and ranking 47th among 250+ teams from Benelux region

EXTRACURRICULAR

PyCon Europe 2024 | 2024

Conference Speaker - 'Performance Engineering for Young Developers'

- Delivered 45-minute technical talk to 280+ attendees covering profiling techniques and optimization case studies; talk received 4.8/5 average rating (146 evaluations) and was selected as top 5 talks
- Published accompanying blog post that reached 15K+ views; created open-source profiling toolkit demonstrated in talk with 340+ GitHub stars

Coding Mentorship Initiative (Imperial + Amsterdam) | 2023 - Present

Senior Mentor - Backend Systems & Databases

- Mentored 8 junior students (first-year undergrads) through capstone projects; 87.5% mentees achieved 1st class marks (up from 62% cohort average) and 6/8 secured internships at tier-1 tech companies
- Conducted weekly 1-hour sessions covering system design, database optimization, and Rust fundamentals; created 14-module curriculum used by 140+ students across both universities

SKILLS

Languages: Python, TypeScript, Rust, JavaScript, SQL, Java, Go

Technologies: React, PostgreSQL, Redis, Kafka, gRPC, GraphQL, Docker, Kubernetes, AWS (EC2, RDS, S3), Git

Tools: VS Code, Git, GitHub, Figma, Datadog, Jira, Docker, Linux, Postman, Chrome DevTools